DTC P0089:00 [SKYACTIV-D 1.5]

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DTC P0089:00	High fuel pressure malfunction in common rail fuel pressure control system
DETECTION CONDITION	• When the following conditions are met, the actual fuel pressure **exceeds 217 Mpa {2,213 kgf/cm^2, 31,473 psi}** for a continuous **3 s**: **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on) — The following DTCs are not detected: • Fuel pressure sensor: P0191:00, P0192:00, P0193:00 **Diagnostic support note** • This is an intermittent monitor (fuel system). • The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• Limits the engine torque or the upper limit of the engine speed. • PCM restricts engine-transaxle integration control. (ATX) • Inhibits engine-stop by operating the i-stop function.
POSSIBLE CAUSE	• Fuel pressure sensor malfunction • Fuel filter clogged • Fuel pipe improper routing (improper connection) • Suction control valve malfunction • Supply pump malfunction • Fuel pressure relief valve malfunction • PCM malfunction
SYSTEM WIRING DIAGRAM	Not applicable

Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD FREEZE FRAME DATA/SNAPSHOT DATA AND DIAGNOSTIC MONITORING TEST RESULTS TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data and DIAGNOSTIC MONITORING TEST RESULTS (fuel system related) on the repair order.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**VERIFY DTCs FOR FUEL PRESSURE SENSOR** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the PENDING CODE/DTC P0191:00, P0192:00 or P0193:00 also present?	Yes	Go to the applicable DTC inspection. (See DTC P0191:00 [SKYACTIV-D 1.5].) (See DTC P0192:00 [SKYACTIV-D 1.5].) (See DTC P0193:00 [SKYACTIV-D 1.5].)
		No	Go to the next step.
4	**INSPECT FUEL PRESSURE SENSOR** • Inspect the fuel pressure sensor. (See FUEL PRESSURE SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the common rail, then go to Step 10. (See COMMON RAIL REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.

STEP	INSPECTION		ACTION
5	**INSPECT FUEL FILTER** • Inspect the fuel filter for clogging. (See FUEL FILTER INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Repair or replace the malfunctioning part according to the inspection results, then go to Step 10. (See FUEL FILTER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) (See FUEL FILTER DISASSEMBLY/ASSEMBLY [SKYACTIV-D 1.5].)
		No	Go to the next step.
6	**INSPECT FUEL PIPE** • Inspect the fuel pipe installation condition. (See FUEL SYSTEM LOCATION INDEX [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Install the fuel pipe properly, then go to Step 10.
		No	Go to the next step.
7	**INSPECT SUCTION CONTROL VALVE** • Inspect the suction control valve. (See SUCTION CONTROL VALVE INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the suction control valve, then go to Step 10. (See SUCTION CONTROL VALVE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
8	**INSPECT SUPPLY PUMP** • Inspect the supply pump. (See SUPPLY PUMP INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the supply pump, then go to Step 10. (See SUPPLY PUMP REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
9	**INSPECT FUEL PRESSURE RELIEF VALVE** • Inspect the fuel pressure relief valve. (See FUEL PRESSURE RELIEF VALVE INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the common rail, then go to the next step. (See COMMON RAIL REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
10	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Drive the vehicle on a flat road surface 5 times under the following conditions. 1. Idle the engine for 20 s. 2. Shift to 2nd gear and rapidly accelerate the vehicle to **35 km/h {22 mph}** and drive the vehicle at a constant speed for **25 s**. 3. Stop the vehicle. 4. Switch the ignition to off and wait for **20 s or more**. 5. Shift to 3rd gear and rapidly accelerate the vehicle to **50 km/h {32 mph}** and drive the vehicle at a constant speed for **50 s**. 6. Idle the engine for **20 s**. • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
11	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.